

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1055

Top Crop 500 kV
August 01, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1055. The Interconnection Customer proposes a 563.0 MW natural gas combined cycle facility interconnecting at the Top Crop 500 kV substation in the Illinois area. The request was received on September 01, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 453 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$49,967,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Wabash 500/230kV Transformer	Richland – Dekalb 500 KV	161.2	1,022	19.5
Montpelier Electric Generating Sta 500 kV/230 kV Transformer	Richland – Dekalb 500 KV	135.3	1,479	14.1
Treat – Chamberlin 230 KV L	El Paso – Ravenna Substation 500 KV line	134.4	633	5.8
Baraga Vlg 500/230 kV Autotransformer	Trego – Beaver 500 KV	133.1	1,689	13.2
Hebo – Barrett 230 KV Line	Winthrop – Hebo 230 KV L	132.2	873	5.7
Darlington Rd Substation – Wayne Trace 100 KV	Richland – Dekalb 500 KV Line	131.5	461	6.0
Madeline – East Methuen 230 KV	Gray'S Flat – East Methuen 230 KV Line	130.4	1,495	9.5
Baraga Vlg – Madeline 230 KV L	Gray'S Flat – East Methuen 230 KV	129.5	963	6.5
Myrtle Pt – George Neal S 230 KV	Trego – Galion Generating Sta Substation 500 KV	129.2	974	6.5
Zona Libre 607 Substation – George Neal South 230 KV	Trego – Galion Generating Sta Substation 500 KV	128.3	1,001	6.5
Zona Libe 6007 – Kampsville Substation 230 KV	Trego – Galion Generating Sta 500 KV	127.4	1,028	6.5
Waldo 500/230 kV Auto	Trego – Beaver 500 KV	127.1	1,248	9.8
Moreno Tap – Kampsville 230 KV	Trego – Galion Generating Sta 500 KV	126.9	1,044	6.5
Ledges – Stoner 230 KV line	El Paso – Ravenna 500 KV L	126.5	618	5.8

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Chamberlin – Involta Substation 230 KV	Ravenna – Hylebos 500 KV	125.4	500	5.3
Burris Pit – James De Young 230 KV	Trego – Beaver 500 KV	124.2	1,392	7.9
E Methuen – Lonetree 500 KV	Muskogee – Montpelier Electric Generating Sta 500 KV	124.1	3,951	9.2
St. Lawrence – James De Young Substation 230 KV	E Methuen – Lonetree 500 KV	124.1	1,621	6.9
Lonetree – Sabanera 500 KV Line	Muskogee – Montpelier Electric Generating Sta 500 KV	123.8	4,315	10.8
Carley Road – James De Young 230 KV	East Methuen 500/230kV Auto	123.3	1,101	7.1
N Cambridge – Atascocita 230 KV	El Paso – Ravenna 500 KV	121.6	1,216	9.1
Corridor – Larson St 230 KV line	Trego – Beaver 500 KV	121.0	1,127	9.9
Norwich 500/1 kV Auto	Barrett – Norwich 500 KV	120.8	1,120	7.0
Gruver – Galeville 230 KV line	Imperial – Eufaula 500 KV	120.7	791	6.6
Baraga Vlg – Gray'S Flat 230 KV	Madeline – E Methuen 230 KV	120.6	1,503	8.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wabash 500/230kV Transformer	Replace transformer with higher-rated unit	\$18,538,000	5% (\$916,000)
Montpelier Electric Generating Station 500/230 kV Auto	Replace transformer with higher-rated unit	\$37,367,000	5% (\$1,845,000)
Chamberlin – Treat 230 KV	Rebuild line with higher-capacity conductor	\$16,606,000	5% (\$818,000)
Baraga Vlg 500/230 kV Autotransformer	Replace transformer with higher-rated unit	\$21,401,000	5% (\$1,058,000)
Barrett – Hebo 230 KV line	Rebuild line with higher-capacity conductor	\$12,167,000	5% (\$602,000)
Wayne Trace – Darlington Rd 100 KV	Rebuild line with higher-capacity conductor	\$2,671,000	5% (\$132,000)
East Methuen – Madeline 230 KV L	Rebuild line with higher-capacity conductor	\$41,818,000	5% (\$2,070,000)
Madeline – Baraga Vlg 230 KV L	Rebuild line with higher-capacity conductor	\$14,966,000	5% (\$740,000)
Myrtle Pt – George Neal S 230 KV	Rebuild line with higher-capacity conductor	\$16,202,000	5% (\$801,000)
Zona Libre 607 Substation – George Neal South 230 KV	Rebuild line with higher-capacity conductor	\$24,201,000	5% (\$1,197,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Zona Libe 6007 – Kampsville Substation 230 KV	Rebuild line with higher-capacity conductor	\$35,933,000	5% (\$1,777,000)
Waldo 500/230 kV Auto	Replace transformer with higher-rated unit	\$20,526,000	5% (\$1,016,000)
Moreno Tap – Kampsville 230 KV line	Rebuild line with higher-capacity conductor	\$47,181,000	5% (\$2,334,000)
Ledges – Stoner 230 KV	Rebuild line with higher-capacity conductor	\$18,130,000	5% (\$893,000)
Chamberlin – Involta Substation 230 KV	Rebuild line with higher-capacity conductor	\$17,472,000	5% (\$863,000)
Burris Pit – James De Young 230 KV Line	Rebuild line with higher-capacity conductor	\$60,707,000	5% (\$3,001,000)
E Methuen – Lonetree 500 KV	Rebuild line with higher-capacity conductor	\$121,133,000	5% (\$5,968,000)
St. Lawrence – James De Young Substation 230 KV	Rebuild line with higher-capacity conductor	\$86,430,000	5% (\$4,271,000)
Lonetree – Sabanera 500 KV Line	Rebuild line with higher-capacity conductor	\$262,426,000	5% (\$12,921,000)
James De Young – Carley Rd 230 KV	Rebuild line with higher-capacity conductor	\$38,343,000	5% (\$1,897,000)
Atascocita – North Cabridge 230 KV Line	Rebuild line with higher-capacity conductor	\$19,562,000	5% (\$964,000)
Corridor – Larson Street 230 KV	Rebuild line with higher-capacity conductor	\$15,770,000	5% (\$781,000)
Norwich 500/1 kV Auto	Replace transformer with higher-rated unit	\$32,626,000	5% (\$1,611,000)
Galeville – Gruver 230 KV	Rebuild line with higher-capacity conductor	\$11,014,000	5% (\$544,000)
Gray'S Flat – Baraga Vlg 230 KV	Rebuild line with higher-capacity conductor	\$19,138,000	5% (\$947,000)
	Total allocated to project		\$49,967,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.